

FINAL EXPLANATION: BIOLOGY GRADE 10

Teacher Key: Exemplar Answers and Marking Guide

FINAL EXPLANATION: BIOLOGY GRADE 10 – SUB-STRAND 3.2 ANIMAL TRANSPORT: FINAL EXPLANATION

Teacher Key: Exemplar Answers (do not give to students)

INSTRUCTIONS GIVEN TO STUDENTS

Driving question: How does the structure of an animal's transport system determine its ability to survive and thrive, and what would happen if Eliud Kipchoge had the circulatory system of a grasshopper? Use evidence from all twelve lessons: your own pulse-rate data, the SA:V cubes, the grasshopper, tilapia, frog and sheep or goat heart work, and the Lesson 1 and Lesson 12 models. Write your answers in the space under each prompt. Use full sentences, scientific terms and, where asked, the Claim-Evidence-Reasoning (CER) pattern. Include labelled sketches where a prompt asks for them.

Part 1: Why Transport Systems Are Needed

Eliud Kipchoge's heart rate falls from over 170 beats per minute to about 37 beats per minute soon after a race. Compare this with your own pulse recovery data after star jumps. (a) Using your SA:V cube investigation (cubes with 1 cm, 2 cm and 3 cm sides), calculate the SA:V ratio of each cube and explain what happens to the ratio as an organism gets bigger. (b) Explain why diffusion alone could not supply Kipchoge's deep muscle cells during a marathon. (c) List four jobs that a transport system does in a large, active animal.

(a) The 1 cm cube has a surface area of 6 cm^2 and a volume of 1 cm^3 , so $\text{SA:V} = 6:1$. The 2 cm cube has a surface area of 24 cm^2 and a volume of 8 cm^3 , so $\text{SA:V} = 3:1$. The 3 cm cube has a surface area of 54 cm^2 and a volume of 27 cm^3 , so $\text{SA:V} = 2:1$. As the cubes got bigger, their volume increased faster than their surface area, so the SA:V ratio fell. A larger body therefore has a smaller share of surface available for exchange compared with the volume that needs supplying. (b) Kipchoge's body is large, and his quadriceps muscle cells lie centimetres from any surface. Diffusion is too slow over such distances, so it could not deliver oxygen and nutrients or remove CO_2 fast enough to sustain marathon pace. A tiny organism like an amoeba has a high SA:V and can exchange substances directly through its membrane. (c) Any four of: delivering oxygen and nutrients to all cells; removing wastes such as CO_2 and urea; transporting hormones; regulating body temperature; circulating immune cells. These needs rise sharply during exercise, as our pulse and breathing data showed.

Part 2: Open and Closed Systems – The Grasshopper

Think about the grasshopper on the school fence. (a) Describe how its circulatory system works, naming the dorsal

(a) The grasshopper has an open circulatory system. The dorsal vessel along its back acts as a simple pump. Haemolymph enters through small openings called ostia and floods the open body cavity (haemocoel), bathing the organs directly. There are no vessels branching to individual organs, and the haemolymph is pale yellow-green, not red. (b) No. Haemolymph does not carry oxygen. Oxygen is delivered directly to cells by a separate tracheal system. (c) The grasshopper is small, so diffusion distances are short,

vessel, ostia, haemocoel and haemolymph. (b) Does haemolymph carry oxygen? Explain how the grasshopper gets oxygen to its cells. (c) Why can the grasshopper survive with this system while a large, fast-moving mammal could not?	and its tracheae supply oxygen without needing the circulatory system. Its open system works at low pressure and cannot direct oxygen-rich fluid to particular muscles on demand. A large, active mammal needs a high-pressure closed system to do this.
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Part 3: Comparing Hearts – Tilapia, Frog and Mammal

Compare the heart and circulation of the tilapia from Lake Victoria, the frog from the Rift Valley wetlands, and a mammal such as Kipchoge. (a) Complete a table with the columns: animal, number of heart chambers, single or double circulation, and what happens to oxygenated and deoxygenated blood. (b) Explain why the tilapia's blood reaches the body with lower pressure than blood in a mammal. (c) Explain why a frog's body tissues receive only partly oxygenated blood, and why this trade-off is acceptable for a frog but not for Kipchoge.	(a) Animal Heart chambers Circulation Blood separation			
	Tilapia	Two (one atrium, one ventricle)	Single, closed	Heart to gills to body to heart; one loop
	Frog	Three (two atria, one ventricle)	Double, but ventricle shared	Oxygenated and deoxygenated blood mix in the single ventricle
	Mammal (Kipchoge)	Four	Double, closed	Complete separation by the septum; no mixing
(b) In the tilapia, blood passes through two sets of capillaries (gills, then body) before returning to the heart. Pressure drops after the gill capillaries and the blood is not pumped again, so delivery to body tissues is less forceful. In a mammal, the left ventricle re-pressurises oxygenated blood before it goes to the body. (c) In the frog, blood from the lungs and skin and blood from the body mix in the single ventricle, so tissues receive partly oxygenated blood. This is acceptable because frogs are ectotherms with lower energy demands. Kipchoge has very high, sustained oxygen demand, so he needs a four-chambered heart that delivers fully oxygenated blood.				

Part 4: The Mammalian Heart and Kipchoge's Recovery

Use your sheep or goat heart observations and your pulse-rate data. (a) Describe two structural features of the mammalian heart and say what each does. (b) Explain what the sino-atrial node (SAN) does and what causes the 'lub-dub' sounds. (c) Kipchoge's heart rate is about 37 beats per	(a) Any two: a thick muscular septum separates the left and right sides so oxygenated and deoxygenated blood do not mix; the left ventricle wall is thicker than the right, giving enough force to pump blood around the whole body; valves between atria and ventricles prevent backflow; coronary vessels on the outside supply the heart muscle itself. (b) The SAN, in the right atrial wall, generates the electrical impulse that starts each heartbeat, so it is the heart's pacemaker. The 'lub' is the closing of the bicuspid and tricuspid valves at the start of ventricular systole; the 'dub' is the closing of the semi-lunar valves at its end. (c) Cardiac output is heart rate multiplied by stroke volume. Because each beat of Kipchoge's trained ventricles pumps about 135 mL, nearly double the roughly 67 mL of an average person, he needs only about 37 beats per minute to deliver the same blood volume per minute at rest. This large stroke volume explains his low resting rate, but it does not by itself explain how fast he recovers. After a race his heart rate falls from over 170 beats per minute to near 37 within minutes, much faster than our own pulse recovered. The speed of this fall is a matter of how the SAN's firing rate is controlled: once exercise stops and the body's demand drops, the SAN can slow
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minute at rest, yet each beat pumps about 135 mL, compared with about 67 mL for an average person. Explain how these figures allow his resting heart rate to be so low. Then use your pulse-rate recovery data and the Lesson 1 curves to describe how his heart rate changes after a race, and suggest what in the control of his heart rate could account for how quickly it returns to rest.	its firing quickly, and the rate settles back to his low resting baseline. Our lessons showed that the SAN sets the rate, so the evidence suggests that fast adjustment of the SAN's rate, together with the low resting baseline set by his large stroke volume, produces his rapid recovery. The details of how that control works were not investigated in our lessons, so this remains a reasoned suggestion rather than something we measured.
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Part 5: Final Explanation (CER)

Answer the driving question using CER. Claim: how does the structure of an animal's transport system determine its ability to survive and thrive? Evidence: give at least three pieces of evidence from different lessons, naming the animals. Reasoning: explain how the evidence supports your claim. Then explain what would happen if Eliud Kipchoge had the circulatory system of a grasshopper. Finally, say how your final model differs from your Lesson 1 model.	Claim: The structure of an animal's transport system determines how well it can meet its oxygen and nutrient demands, so it sets the level of activity the animal can sustain. Evidence: (1) The grasshopper has an open system with haemolymph that does not carry oxygen, and tracheae deliver oxygen directly, which suits its small size and low demand. (2) The tilapia has a closed, single circulation with a two-chambered heart, and pressure drops after the gills. (3) The frog's three-chambered heart mixes oxygenated and deoxygenated blood. (4) Kipchoge's four-chambered heart with double circulation keeps the blood fully separated, and his large stroke volume (about 135 mL) lets his heart rate be about 37 beats per minute at rest. Reasoning: Across the animals, circulation becomes more complex: open to closed, single to double, and mixed to fully separated blood. Each step improves oxygen delivery per heartbeat and matches the animal's energy demand. Kipchoge's size and muscle demand are very high, so he needs the most efficient system. If he had a grasshopper's system, he is far too large for diffusion and low-pressure haemolymph to supply his leg muscles, and haemolymph does not carry oxygen. The system could not direct oxygen-rich fluid to working muscles on demand, so his muscles would fail quickly at marathon pace. Without a high-pressure closed system with a powerful heart and SAN, he would also lack the means to meet his body's demand and return rapidly to his resting rate of about 37 beats per minute. Model change: My Lesson 1 sketch was a rough guess at a transport system. My final model shows open and closed systems, the heart chambers of each animal, the path of blood in the double circuit, and the SAN, with evidence labels.
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Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Explaining why transport systems are needed	Links SA:V ratio, diffusion distance and the functions of transport accurately, using the cube and pulse data.	Explains that large animals need transport and names some functions, with limited use of SA:V evidence.	States that animals need transport without a clear reason or evidence.
Comparing transport systems across animals	Accurately compares grasshopper, tilapia, frog and mammal on open or closed, single or double, and mixing of	Describes most systems correctly with a few gaps or errors.	Describes few systems, or confuses their features.

	blood, using correct terms.		
Use of evidence and CER reasoning	Gives at least three accurate pieces of evidence from different lessons and links each clearly to the claim.	Gives two or three pieces of evidence with partial reasoning.	Gives little or vague evidence; claim is not supported.
Applying ideas to Kipchoge and the grasshopper scenario	Explains fully why a grasshopper system would fail Kipchoge, covering size, pressure, oxygen carrying and recovery, using the heart rate and stroke volume data.	Explains that the system would not supply enough oxygen, with some reasons.	Gives an unsupported or incorrect prediction.
Scientific communication and model reflection	Uses scientific vocabulary precisely and clearly describes how the final model differs from the Lesson 1 model.	Uses mostly correct vocabulary and names some changes to the model.	Vocabulary is often incorrect and model changes are not described.

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